



ch



Solve the following system:

$$x_1 - 2x_2 + x_3 = 0$$

$$2x_2 - 8x_3 = 8$$

$$5x_1 - 5x_3 = 10$$

Augmented  
matrix

Step 1

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$$

Step 2

$$R_3 = R_3 - 5 \times R_1$$

$$R_3 = (5, 0, -5, 10) - 5(1, -2, 1, 0)$$

$$\downarrow$$

$$= (-5, +10, -5, 0)$$

$$R_3 (0 \ 10 \ -10 \ 10)$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 10 & -10 & 10 \end{bmatrix}$$

Step 3

$$R_2 = \frac{1}{2} \times R_2$$

$$\begin{cases} x + 2y - z = 3 \\ 2x - y + 3z = 5 \\ 3x + 6y - 3z = 10 \end{cases}$$

Pivot  $\rightarrow$

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 5 \\ 3 & 6 & -3 & 10 \end{array} \right]$$

$$R_2 = -2(R_1) + R_2$$

$$-2(1, 2, -1, 3) + (2, -1, 3, 5)$$

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -1 \\ (3 & 6 & -3) & 10 \end{array} \right]$$

$$R_3 = -3(R_1) + R_3$$

$$-3(1, 2, -1, 3) + (3, 6, -3, 10)$$

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -1 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

# Lecture 2

then it is in **reduced echelon form** (or **reduced row echelon form**):

4. The leading entry in each nonzero row is 1.  
5. Each leading 1 is the only nonzero entry in its column.

2

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

Echelon Form

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Reduced Echelon Form

**Echelon Form is not unique**

A nonzero matrix can have **more than one** equivalent matrices in **Echelon Forms**

**Reduced Echelon Form is unique**

A nonzero matrix can have **only one** equivalent matrix in **Reduced Echelon Forms**

3

02:38 / 14:02

# lecture 3

$$u = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ \& } v = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

find  $4u_A$ ,  $-3v_B$  and  $4u_A + (-3)v_B$

$$4 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}_B$$

$$-3 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix}$$

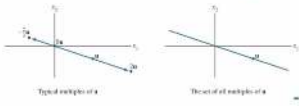
$$\begin{bmatrix} -2 \\ 7 \end{bmatrix}$$

Length ?

Example

Let  $u = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$ . Display the vectors  $u$ ,  $2u$ , and  $-\frac{2}{3}u$  on a graph.

$2u = \begin{bmatrix} 12 \\ -4 \end{bmatrix}$ , and  $-\frac{2}{3}u = \begin{bmatrix} -4 \\ \frac{4}{3} \end{bmatrix}$



the length of the arrow for  $cu$  is  $|c|$  \* the length of the arrow for  $u$

The length of the arrow for  $u$  with entries  $\begin{bmatrix} a \\ b \end{bmatrix}$  is calculate  $\sqrt{a^2 + b^2}$ , which

is the length of the line segment between the points  $(0, 0)$  and  $(a, b)$

Vectors in  $\mathbb{R}^3$

- $3 \times 1$  column matrices with three entries

### Example

Let  $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ ,  $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ , and  $\mathbf{b} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$

Determine whether  $\mathbf{b}$  can be generated as a linear combination of  $\mathbf{a}_1$  and  $\mathbf{a}_2$ .

**Solution:** If  $\mathbf{b}$  is a linear combination of  $\mathbf{a}_1$  and  $\mathbf{a}_2$ , then *or span*

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b} \rightarrow \text{augmented matrix } [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{b}]$$

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\xrightarrow{\substack{R_2=R_2+2R_1 \\ R_3=R_3+5R_1}} \quad \xrightarrow{R_2=1/9 R_2} \quad \xrightarrow{\substack{R_3=R_3-16R_2 \\ R_1=R_1-2R_2}}$

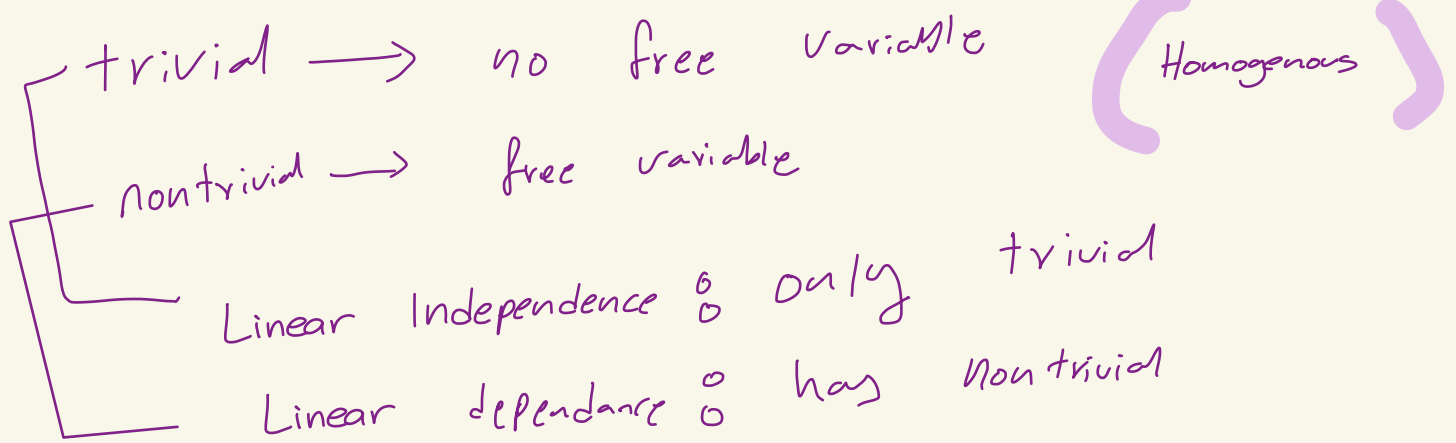
**Solution:**  $x_1 = 3$  and  $x_2 = 2$

Since the system has a solution, then  $\mathbf{b}$  can be generated as a linear combination of  $\mathbf{a}_1$  and  $\mathbf{a}_2$ , and the combination is:

$$3\mathbf{a}_1 + 2\mathbf{a}_2 = \mathbf{b}$$

$$3 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

ch1



onto  $\rightarrow$  consistent

one to one  $\rightarrow$  no free variable

independent

if its  
not a  
zero  
vector

Dependent

if its  
Zero  
vector

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

# transformation

**EXAMPLE 1** Let  $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$ ,  $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ ,  $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$ ,  $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ , and define a transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  by  $T(\mathbf{x}) = A\mathbf{x}$ , so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

a. Find  $T(\mathbf{u})$ , the image of  $\mathbf{u}$  under the transformation  $T$ .

Compute

$$T(\mathbf{u}) = A\mathbf{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

find  
 $T(\mathbf{u})$

find  
 $\downarrow$   
the solution

multiply

b. Find an  $\mathbf{x}$  in  $\mathbb{R}^2$  whose image under  $T$  is  $\mathbf{b}$ .

Solve  $T(\mathbf{x}) = \mathbf{b}$  for  $\mathbf{x}$ . That is, solve  $A\mathbf{x} = \mathbf{b}$ , or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

$x_1 = 1.5$   $x_2 = -0.5$

Solution  $\begin{cases} x_1 = 1.5 \\ x_2 = -0.5 \end{cases}$   $\mathbf{x} = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$

check  $A\mathbf{x} = \mathbf{b}$   $\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} = \mathbf{b}$

Therefore, the vector whose image is  $\mathbf{b}$  is the vector  $\mathbf{x} = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$

c. Is there more than one  $\mathbf{x}$  whose image under  $T$  is  $\mathbf{b}$ ?

step 1:

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

step 2

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix}$$

step 3 RREF

Solution  $x_1 = 1.5$   
 $x_2 = -0.5$

c. Is there more than one  $\mathbf{x}$  whose image under  $T$  is  $\mathbf{b}$ ?

We need to find whether  $A\mathbf{x} = \mathbf{b}$  has a unique solution

We have already done that in part (b) above

either from the echelon form:

$$\begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

no false condition and no free variable.

Therefore, it has a unique solution.

OR from the final solution

$$\mathbf{x} = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$$

So, there is exactly one  $\mathbf{x}$  whose image is  $\mathbf{b}$

d. Determine if  $\mathbf{c}$  is in the range of the transformation  $T$ .

$$\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

asking if the system  $A\mathbf{x} = \mathbf{c}$  is consistent.

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

$$0 = -35$$

The third equation,  $0 = -35$ , shows that the system is inconsistent. So  $\mathbf{c}$  is not in the range of  $T$ .

Example: shear transformation

Consistent?

in Range

not?

not in Range

$$0 = -35$$

inconsistent

Not in Range

**EXAMPLE 5** Define a linear transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

Find the images under  $T$  of  $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ ,  $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$  and  $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ .

**SOLUTION**

$$T(\mathbf{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad T(\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

$$T(\mathbf{u} + \mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

$T$  rotates  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{u} + \mathbf{v}$  counterclockwise about the origin through  $90^\circ$ .

In this example, the angle is  $90^\circ$  degrees

**EXAMPLE 2** Find the standard matrix  $A$  for the dilation transformation  $T(\mathbf{x}) = 3\mathbf{x}$ , for  $\mathbf{x}$  in  $\mathbb{R}^2$ .

1. This transformation multiplies each vector input by 3

2. Since the domain is  $\mathbb{R}^2$  (given), we use the identity matrix  $I_2$  which is  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

3. Apply the transformation on each column of  $I_2$ , let's call them  $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  and  $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

we get these results:  $T(\mathbf{e}_1) = 3\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$  and  $T(\mathbf{e}_2) = 3\mathbf{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

4. Each result is a column in the matrix  $A$ :  $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

the standard matrix of linear transformation

*Handwritten notes:*  $I \in \mathbb{R}^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Existence and Uniqueness Questions

Ch2

How to add  
& multiply  
matrix

Warnings:

1. In general,  $AB \neq BA$ .
2. The cancellation laws do *not* hold for matrix multiplication. That is, if  $AB = AC$ , then it is *not* true in general that  $B = C$ .
3. If a product  $AB$  is the zero matrix, you *cannot* conclude in general that either  $A = 0$  or  $B = 0$ .

$3 \times 5 = 5 \times 3$  ✓

**Example**

Let  $A = \begin{bmatrix} 3 & -3 \\ -4 & 6 \end{bmatrix}$ ,  $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$ , and  $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$ .

$AB = \begin{bmatrix} 1 & 7 \\ -2 & 14 \end{bmatrix}$

$AC = \begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix}$

$AB \neq AC$

**Example**

Let  $A = \begin{bmatrix} 5 & -6 \\ -1 & 2 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$ , and  $C = \begin{bmatrix} 4 & 8 \\ 2 & 4 \end{bmatrix}$ .

$AB = 0$

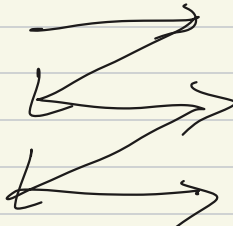
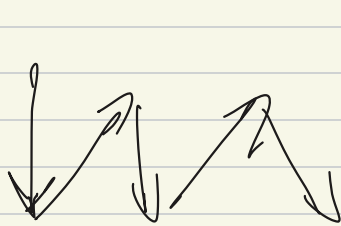
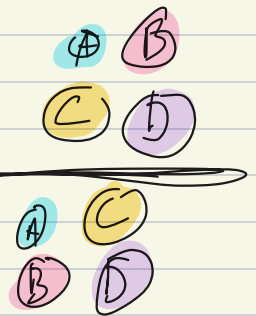
$AC = 0$

transpose

$B^T$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$\begin{matrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{matrix}$$



ex

$$C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{bmatrix}$$

$$C^T = \begin{matrix} 1 & -3 \\ 1 & 5 \\ 1 & -2 \\ 1 & 7 \end{matrix}$$

ex

$$B = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}$$

$$B^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}$$

---

note

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(rA)^T = rA^T$$

$$(AB)^T = B^T A^T$$

# Ch2 (Summary)

$$B^T \quad B = \begin{bmatrix} A & B \\ C & D \\ e & f \end{bmatrix} \rightarrow B^T = \begin{bmatrix} A & C & e \\ B & D & f \end{bmatrix}$$

$$I \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(identity)

$$\mathbb{R}^2 \rightarrow 2 \times 2$$
$$\mathbb{R}^3 \rightarrow 3 \times 3$$

Inverse

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

if  $ad-bc \neq 0$  its invertable

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

if  $ad-bc = 0$  its **Not** invertable

$$A^{-1}A = I$$

$$AA^{-1} = I$$

Lecture2\_Chapter2

### The inverse of a Matrix

An  $n \times n$  matrix  $A$  is said to be **invertible** if there is an  $n \times n$  matrix  $C$  such that

$$CA = I \quad \text{and} \quad AC = I$$

where  $I = I_n$ , the  $n \times n$  identity matrix. In this case,  $C$  is an **inverse** of  $A$ .

$$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I$$

**EXAMPLE 1** If  $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$  and  $C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$ , then  $2 \times 2$   $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  and  $3 \times 3$   $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$AC = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad CA = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus  $C = A^{-1}$ .

The inverse of a  $2 \times 2$  Matrix

$$X = A^{-1}b$$

**EXAM**

**SOLUTION:** Since  $\det(A) = (6)(-4) - (-2) \neq 0$ ,  $A$  is invertible, and

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 6 & -4 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -1/2 \end{bmatrix}$$

### The inverse of a Matrix

If  $A$  is an invertible  $n \times n$  matrix, then for each  $b$  in  $\mathbb{R}^n$ , the equation  $Ax = b$  has the unique solution  $x = A^{-1}b$ .

**EXAMPLE 4** Use the inverse of the matrix  $A$  in Example 2 to solve the system

$$\begin{cases} 3x_1 + 4x_2 = 3 \\ 5x_1 + 6x_2 = 7 \end{cases} \quad \begin{bmatrix} 3 \\ 7 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

**SOLUTION:** This system is equivalent to  $Ax = b$ , so

$$x = A^{-1}b = \begin{bmatrix} -3 & 2 \\ 5/2 & -1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

### The inverse of a Matrix

a. If  $A$  is an invertible matrix, then  $A^{-1}$  is invertible and

$$(A^{-1})^{-1} = A$$

b. If  $A$  and  $B$  are  $n \times n$  invertible matrices, then so is  $AB$ , and the inverse of  $AB$  is the product of the inverses of  $A$  and  $B$  in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

c. If  $A$  is an invertible matrix, then so is  $A^T$ , and the inverse of  $A^T$  is the transpose of  $A^{-1}$ . That is,

$$(A^T)^{-1} = (A^{-1})^T$$

- if its  $3 \times 3$

use this to find  
inverse

$$[A \ I]$$

ex:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left[ \begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right]$$

Put in one  
matrix  
then solve in RRef

EXAMPLE 7 Find the inverse of the matrix  $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ , if it exists.

SOLUTION

$[A \ I] = \left[ \begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right] \sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & 8 & 4 & -3 & 1 \end{array} \right]$

$\sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 14 & 7 & -3 & 1 \end{array} \right] \sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 5 & -1 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 14 & 7 & -3 & 1 \end{array} \right]$

$\sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 5 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 14 & 7 & -3 & 1 \end{array} \right] \sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 5 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 1/2 & -3/14 & 1/14 \end{array} \right]$

check

$A^{-1} = \begin{bmatrix} -3 & 5 & -1 \\ -1 & 2 & -1 \\ 1/2 & -3/14 & 1/14 \end{bmatrix}$

Characterization of Invertible Matrices

**EXAMPLE 1** Use the Invertible Matrix Theorem to decide if  $A$  is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

**SOLUTION**

$$A \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix} \quad \checkmark$$

So  $A$  has three pivot positions and hence is invertible, by the Invertible Matrix Theorem, statement (c). ■

$$Ly = b$$

$$ux = y$$

$$Lb = y$$

$$uy = x$$

**Factorization (two)**

Example:  $A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = LU$

Use this LU factorization of  $A$  to solve  $Ax = b$ , where  $b = \begin{bmatrix} 2 \\ 7 \\ 8 \\ 11 \end{bmatrix}$ .

**Step 1: Find  $y$**   $Ly = b$

Augmented matrix  $[L \ b] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad y = \begin{bmatrix} -9 \\ -4 \\ 5 \\ 1 \end{bmatrix}$

The forward algorithm to get solution form:  
Submatrix 1:  $R2 = R2 + R1 \rightarrow R3 = R3 - 2R1 \rightarrow R4 = R4 + 3R1$   
Submatrix 2:  $R3 = R3 - R2 \rightarrow R4 = R4 - R2$   
Submatrix 3:  $R4 = R4 - R3$

**Step 2: Find  $x$**   $Ux = y$

Augmented matrix  $[U \ y] = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \quad x = \begin{bmatrix} 3 \\ 4 \\ 5 \\ -1 \end{bmatrix}$

The backward algorithm to get reduced echelon form:  
C4:  $R4 + R1 \rightarrow R3 + R2 + R4 \rightarrow R2 = R2 - 2R1 \rightarrow R1 = R1 - 2R4 \rightarrow C4$   
C3:  $R3 - R2 \rightarrow R2 = R2 + R3 \rightarrow R3 = R3 + 2R1 \rightarrow C3$   
C2:  $R2 = R2 / 2 \rightarrow R1 = R1 + 7R2 \rightarrow C2$   
C1:  $R1 = R1 / 3 \rightarrow C1$

Total arithmetic operations in both Step 1 and 2 is around **28 operations**, (not counting LU factorization)

If we go the ordinary way (just in one step), it takes **62 operations**!!

But how do we get  $U$  and  $L$ ? See the next slide .....

**An LU Factorization Algorithm**

ALGORITHM FOR AN LU FACTORIZATION  
1. Reduce  $A$  to an echelon form  $U$  by a sequence of row replacement operations.

Column Space

The column space is the same as the set of all linear combinations of the columns of the matrix.

EXAMPLE 4 Let  $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$  and  $b = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$ . Determine whether  $b$  is in the column space of  $A$ .

SOLUTION: The vector  $b$  is a linear combination of the columns of  $A$  if and only if  $b$  can be written as  $Ax$  for some  $x$ , that is, if and only if the equation  $Ax = b$  has a solution. Row reducing the augmented matrix  $[A \ b]$ :

$$\left[ \begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -5 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

we conclude that  $Ax = b$  is consistent and  $b$  is in  $\text{Col } A$ .

Subspace

- ① Add  $b$  to to matrix
- ② Ref
- ③ its consistent? yes
- ④  $b$  is in  $\text{col } A$

# null space

$A\mathbf{x} = \mathbf{0}$

1. Write the solution in parametric vector form

2. Take the pivot columns from the reduced echelon form) as the basis

3. Take the vectors of  $\mathbf{x}$  as the basis

**EXAMPLE 6** Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

**SOLUTION** First, write the solution of  $A\mathbf{x} = \mathbf{0}$  in parametric vector form:

$$[A \ \mathbf{0}] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

The general solution is  $x_1 = 2x_2 + x_4 - 3x_5$ ,  $x_3 = -2x_4 + 2x_5$ , with  $x_2$ ,  $x_4$ , and  $x_5$  free.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$   $\uparrow$   $\uparrow$   
 $\mathbf{u}$   $\mathbf{v}$   $\mathbf{w}$

$= x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$  (1)

①  $[A \ \mathbf{0}]$  (just add 0 column)

$$\begin{array}{c} x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \\ \left[ \begin{array}{cccccc} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right], \end{array}$$

①  $x_1 - 2x_2 - x_4 + 3x_5 = 0$

$x_3 + 2x_4 - 2x_5 = 0$

$0 = 0$

$$\textcircled{2} \quad X_1 = 2x_2 + x_4 - 3x_5$$

$$X_2 = \text{free}$$

$$X_3 = -2x_4 + 2x_5$$

$$X_4 = \text{free}$$

$$X_5 = \text{free}$$

$\textcircled{3}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

Find Basis of  
column space

Find a basis for the column space of the matrix

$$A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 9 & -1 & 5 \\ 2 & -6 & 4 & -3 \\ -4 & 12 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\{ \begin{bmatrix} 1 \\ -3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix} \}$

$$\left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix} \right\}$$

Order, Dimension, Rank, Nullity  
Column Space, Null space of a matrix.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{bmatrix}$$

$R^1 \ R^2 \ R^3$

Row-Echelon Form

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Reduced-Row Echelon Form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1. Order of a matrix = number of elements.  
A has order 9.

②. Dimension of A =  $3 \times 3$  (rows  $\times$  columns)

③. Rank of A = 2 (# of linearly independent columns)

④. Nullity of A = 1 (# of zero columns in RREF)

⑤. Column space =  $\text{span}\{\text{columns of A}\} = \text{span}\left\{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}\right\}$

⑥. Null space =  $\text{span}\{\text{zero columns in RREF}\}$

\* Rank + Nullity = # of columns.

**EXAMPLE 1** Let  $v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ ,  $v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ ,  $x = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$ , and  $B = \{v_1, v_2\}$ . Then  $B$  is a basis for  $H = \text{Span}\{v_1, v_2\}$  because  $v_1$  and  $v_2$  are linearly independent. Determine if  $x$  is in  $H$ , and if it is, find the coordinate vector of  $x$  relative to  $B$ .

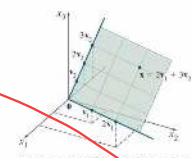
**SOLUTION** If  $x$  is in  $H$ , then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

The scalars  $c_1$  and  $c_2$ , if they exist, are the  $B$ -coordinates of  $x$ . Row operations show that

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus  $c_1 = 2$ ,  $c_2 = 3$ , and  $[x]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ . The basis  $B$  determines a "coordinate system" on  $H$ , which can be visualized by the grid shown in Figure 1.



**FIGURE 1** A coordinate system on a plane  $H$  in  $\mathbb{R}^3$ .

The Dimension of a Subspace

① Ref

the answer is  
coordinate

Rank is Number  
Pivot

$$\begin{array}{ccc|c} 1 & -3 & 3 & \\ 2 & 3 & -1 & \\ 4 & -3 & -1 & \end{array} \quad \left| \quad L = \begin{bmatrix} 1 & 0 & 0 \\ ? & 1 & 0 \\ ? & ? & 1 \end{bmatrix} \right.$$

$-2R_1 + R_2 \rightarrow R_2$

$$\begin{array}{ccc|c} 1 & -3 & 3 & \\ 0 & 9 & -7 & \\ 4 & -3 & -1 & \end{array} \quad \left| \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ ? & ? & 1 \end{bmatrix} \right.$$

$-4R_1 + R_3 \rightarrow R_3$

$$\begin{array}{ccc|c} 1 & -3 & 3 & \\ 0 & 9 & -7 & \\ 0 & 9 & -13 & \end{array} \quad \left| \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & ? & 1 \end{bmatrix} \right.$$

$-R_2 + R_3 \rightarrow R_3$

$$\begin{array}{ccc|c} 1 & -3 & 3 & \\ 0 & 9 & -7 & \\ 0 & 0 & -6 & \end{array} \quad \left| \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \right.$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & -3 & 3 \\ 0 & 9 & -7 \\ 0 & 0 & -6 \end{bmatrix}}_U = \underbrace{\begin{bmatrix} 1 & -3 & 3 \\ 2 & 3 & -1 \\ 4 & -3 & -1 \end{bmatrix}}_A$$

$A = LU$

$$Ax = b$$

$$LUx = b$$

$$Ux = d$$

$$\Rightarrow Ld = b$$

$$\begin{array}{rcl} x - 3y + 3z & = & -4 \\ 2x + 3y - z & = & 15 \\ 4x - 3y - z & = & 19 \end{array}$$

$$\underbrace{\begin{bmatrix} 1 & -3 & 3 \\ 2 & 3 & -1 \\ 4 & -3 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_x = \underbrace{\begin{bmatrix} -4 \\ 15 \\ 19 \end{bmatrix}}_b$$

$\Rightarrow Ld = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 15 \\ 19 \end{bmatrix}$$

$d_1 = -4$   
 $2d_1 + d_2 = 15 \Rightarrow d_2 = 23$   
 $4d_1 + d_2 + d_3 = 19 \Rightarrow d_3 = 12$

$Ux = d$

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 9 & -7 \\ 0 & 0 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ 23 \\ 12 \end{bmatrix}$$

$-6z = 12 \Rightarrow z = -2$   
 $9y - 7z = 23 \Rightarrow y = 1$   
 $x - 3y + 3z = -4 \Rightarrow x = 5$

# Ch3 Lesson 1

Using  
co-factor extension

$$C_{ij} = (-1)^{i+j} A_{ij} C_{ij}$$

$$A = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$(-1)^{3+1} (0) \begin{bmatrix} \phantom{0} \end{bmatrix} + (-1)^{3+2} (-2) \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + (-1)^{3+1} (0) \begin{bmatrix} \phantom{0} \end{bmatrix}$$

$$(-1)(-2)$$

$$(2) \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

$$1 \times -1 - 2 \times 0$$

$$-1 - 0$$

$$2(-1) = -2$$

$$\begin{bmatrix} 1 & 4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$$

Reduce

$$\begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{bmatrix}$$

compute det

$$\rightarrow -(1)(3)(-5)$$

$$= 15$$

# ch3 Lesson 2

A square matrix  $A$  is invertible if and only if  $\det A \neq 0$ .

**SOLUTION** Add 2 times row 1 to row 3 to obtain

$$\det A = \det \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} = 0$$

## Column Operations

**THEOREM 5** If  $A$  is an  $n \times n$  matrix, then  $\det A^T = \det A$ .

Therefore, we can perform operations on the columns of a matrix in a way that is analogous to the row operation, and the determinant won't be affected.

## Determinants and Matrix Products

**THEOREM 6** **Multiplicative Property**

If  $A$  and  $B$  are  $n \times n$  matrices, then  $\det AB = (\det A)(\det B)$ .

**Warning:** A common misconception is that Theorem 6 has an analogue for sums of matrices. However,  $\det(A + B)$  is not equal to  $\det A + \det B$ , in general.

## Cramer's Rule

For any  $n \times n$  matrix  $A$  and any  $\mathbf{b}$  in  $\mathbb{R}^n$ , let  $A_i(\mathbf{b})$  be the matrix obtained from  $A$  by replacing column  $i$  by the vector  $\mathbf{b}$ .

$$A_i(\mathbf{b}) = [\mathbf{a}_1 \ \cdots \ \mathbf{b} \ \cdots \ \mathbf{a}_n]$$

$\uparrow$   
col  $i$

**THEOREM 7**

**Cramer's Rule**

Let  $A$  be an invertible  $n \times n$  matrix. For any  $\mathbf{b}$  in  $\mathbb{R}^n$ , the unique solution  $\mathbf{x}$  of  $A\mathbf{x} = \mathbf{b}$  has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

$$\frac{\det A_i(\mathbf{b})}{\det A}$$

**EXAMPLE 1** Use Cramer's rule to solve the system

$$\begin{aligned} 3x_1 - 2x_2 &= 6 \\ -5x_1 + 4x_2 &= 8 \end{aligned}$$

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\mathbf{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\mathbf{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

**SOLUTION** View the system as  $A\mathbf{x} = \mathbf{b}$ . Using the notation introduced above,

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\mathbf{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\mathbf{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since  $\det A = 2$ , the system has a unique solution. By Cramer's rule,

$\det A \neq 0$ .  
Therefore,  $A$  is invertible

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{24 + 30}{2} = 27$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$ad - bc \text{ det}$$

$$A = \begin{bmatrix} 1 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$$

if its  
3x3

# ch3 Lesson 3

## A Formula for $A^{-1}$

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{ln} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

Steps:

1. Calculate cofactor at position  $(i, j)$  of the original matrix  $A$ .
2. Assign result to position  $(j, i)$  (opposite order of  $i$  and  $j$ ) in the adjugate matrix ( $\text{adj } A$ )
3. Repeat for all entries in  $A$  and  $\text{adj } A$
4. Multiply scalar  $1/\det A$  by  $\text{adj } A$

The matrix of cofactors on the right side of (4) is called the **adjugate** (or **classical adjoint**) of  $A$ , denoted by  $\text{adj } A$ . (The term *adjoint* also has another meaning in advanced texts on linear transformations.) The next theorem simply restates (4).

### THEOREM 8

#### An Inverse Formula

Let  $A$  be an invertible  $n \times n$  matrix. Then

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

### EXAMPLE 3

Find the inverse of the matrix  $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$ .

**SOLUTION** The nine cofactors are

$$\begin{aligned} C_{11} &= + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, & C_{12} &= - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, & C_{13} &= + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5 \\ C_{21} &= - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, & C_{22} &= + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, & C_{23} &= - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7 \\ C_{31} &= + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, & C_{32} &= - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, & C_{33} &= + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3 \end{aligned}$$

The adjugate matrix is the *transpose* of the matrix of cofactors. [For instance,  $C_{12}$  goes in the  $(2, 1)$  position.] Thus

$$\text{adj } A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

$$\det A = 14 \text{ and}$$

$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -1/2 & -3/14 \end{bmatrix}$$

$$\frac{1}{\det} \left[ \begin{array}{ccc} / & / & / \end{array} \right]$$

check calculator

$$A^{-1} =$$

**EXAMPLE 4** Calculate the area of the parallelogram determined by the points

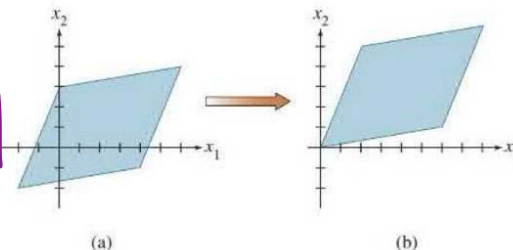
$(-2, -2)$ ,  $(0, 3)$ ,  $(4, -1)$ , and  $(6, 4)$ . See Figure 5(a).

**SOLUTION** First translate the parallelogram to one having the origin as a vertex. For example, subtract the vertex  $(-2, -2)$  from each of the four vertices. The new parallelogram has the same area, and its vertices are  $(0, 0)$ ,  $(2, 5)$ ,  $(6, 1)$ , and  $(8, 6)$ . See Figure 5(b). This parallelogram is determined by the columns of

$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$$

Since  $|\det A| = |-28|$ , the area of the parallelogram is 28.

$$|\det A| = \begin{vmatrix} 2 & 6 \\ 5 & 1 \end{vmatrix} = 2 \times 1 - 6 \times 5 = 2 - 30 = -28 = 28$$



**FIGURE 5** Translating a parallelogram does not change its area.

#### THEOREM 10

Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the linear transformation determined by a  $2 \times 2$  matrix  $A$ . If  $S$  is a parallelogram in  $\mathbb{R}^2$ , then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

If  $T$  is determined by a  $3 \times 3$  matrix  $A$ , and if  $S$  is a parallelepiped in  $\mathbb{R}^3$ , then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

**Example.** Let  $S$  be the parallelogram determined by the vectors  $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$  and  $\mathbf{b}_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ , and

$\det A = \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} = 1 \times 2 - (-1 \times 0) = 2$ . Compute the area of the image of  $S$  under the mapping  $\mathbf{x} \mapsto A\mathbf{x}$ .

The area of  $S$  is  $|\det \begin{bmatrix} 1 & 5 \\ 3 & 1 \end{bmatrix}| = 14$ , and  $\det A = 2$ . By Theorem 10, the area of the image of  $S$  under the mapping  $\mathbf{x} \mapsto A\mathbf{x}$  is

$$|\det A| \cdot \{\text{area of } S\} = 2 \cdot 14 = 28$$

$$2 \times 14 = 28$$

transformation multiplied by the det of  $A$

The volume after transformation is equal to the volume before transformation multiplied by the det of  $A$

# Ch 4 Lesson 1

$$\begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$$

**EXAMPLE 1** Let  $A$  be the matrix in (2), and let  $\mathbf{u} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$ . Determine if  $\mathbf{u}$  belongs to the null space of  $A$ .

**SOLUTION** To test if  $\mathbf{u}$  satisfies  $A\mathbf{u} = \mathbf{0}$ , simply compute

$$A\mathbf{u} = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 - 9 + 4 \\ -25 + 27 - 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus  $\mathbf{u}$  is in  $\text{Nul } A$ .

*nul*

Just multiply

$$\begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$$

$$5 - 9 + 4 = 0$$

$$-25 + 27 - 2 = 0$$

**EXAMPLE 3** Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

**SOLUTION** The first step is to find the general solution of  $A\mathbf{x} = \mathbf{0}$  in terms of free variables. Row reduce the augmented matrix  $[A \ \mathbf{0}]$  to reduced echelon form in order to write the basic variables in terms of the free variables:

$$\left[ \begin{array}{ccccc|c} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right], \quad \begin{array}{l} x_1 - 2x_2 - x_4 + 3x_5 = 0 \\ x_3 + 2x_4 - 2x_5 = 0 \\ 0 = 0 \end{array}$$

The general solution is  $x_1 = 2x_2 + x_4 - 3x_5$ ,  $x_3 = -2x_4 + 2x_5$ , with  $x_2$ ,  $x_4$ , and  $x_5$  free. Next, decompose the vector giving the general solution into a linear combination of vectors where the weights are the free variables. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$$

$$\begin{array}{l} x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \\ x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \end{array}$$

Reduce



Every linear combination of  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  is an element of  $\text{Nul } A$  and vice versa. Thus  $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$  is a spanning set for  $\text{Nul } A$ .

**EXAMPLE 3** Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\left[ \begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right], \quad x_1$$

The general solution is  $x = 2x_2 + x_3 - 2x_4 - 3x_5$

Switch Signs

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -x_1 + 2x_2 + x_4 - 3x_5 \\ x_2 \\ +x_3 - 2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

$$x_2 \begin{bmatrix} +2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} +1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ +2 \\ 0 \\ 1 \end{bmatrix}$$

**EXAMPLE 4** Find a matrix  $A$  such that  $W = \text{Col } A$ .

$$W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : a, b \text{ in } \mathbb{R} \right\}$$

**SOLUTION** First, write  $W$  as a set of linear combinations.

$$W = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a, b \text{ in } \mathbb{R} \right\} = \text{Span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Second, use the vectors in the spanning set as the columns of  $A$ . Let  $A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$ .

Then  $W = \text{Col } A$ , as desired.

$$a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$$

## The Contrast Between Nul A and Col A

**EXAMPLE 6** Let

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$$

Rows  
3 x 4  
Column Space  
null space

- If the column space of  $A$  is a subspace of  $\mathbb{R}^k$ , what is  $k$ ?
- If the null space of  $A$  is a subspace of  $\mathbb{R}^k$ , what is  $k$ ?

**SOLUTION**

- The columns of  $A$  each have three entries, so  $\text{Col } A$  is a subspace of  $\mathbb{R}^k$ , where  $k = 3$ .
- A vector  $\mathbf{x}$  such that  $A\mathbf{x}$  is defined must have four entries, so  $\text{Nul } A$  is a subspace of  $\mathbb{R}^k$ , where  $k = 4$ . ■

nul A

1. Multiply & Add

2. Result = 0 ?

if not  
 $A\mathbf{x} \neq 0$   
so its not  
nul A

**EXAMPLE 8** With  $A$  as in Example 6, let  $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$ .

Is  $\mathbf{u}$  a solution for  $A\mathbf{x} = \mathbf{0}$ ?

- Determine if  $\mathbf{u}$  is in  $\text{Nul } A$ . Could  $\mathbf{u}$  be in  $\text{Col } A$ ?
- Determine if  $\mathbf{v}$  is in  $\text{Col } A$ . Could  $\mathbf{v}$  be in  $\text{Nul } A$ ?

**SOLUTION**

Is  $\mathbf{v}$  in the span?

- An explicit description of  $\text{Nul } A$  is not needed here. Simply compute the product  $A\mathbf{u}$ .

$$A\mathbf{u} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Obviously,  $\mathbf{u}$  is not a solution of  $A\mathbf{x} = \mathbf{0}$ , so  $\mathbf{u}$  is not in  $\text{Nul } A$ . Also, with four entries,  $\mathbf{u}$  could not possibly be in  $\text{Col } A$ , since  $\text{Col } A$  is a subspace of  $\mathbb{R}^3$ .

- Reduce  $[A \ \mathbf{v}]$  to an echelon form.

$$[A \ \mathbf{v}] = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

At this point, it is clear that the equation  $A\mathbf{x} = \mathbf{v}$  is consistent, so  $\mathbf{v}$  is in  $\text{Col } A$ . With only three entries,  $\mathbf{v}$  could not possibly be in  $\text{Nul } A$ , since  $\text{Nul } A$  is a subspace of  $\mathbb{R}^4$ . ■

$\text{Col } A = [A]_{\text{then Ref}}$

not

Col A

1. Reduce

as v as the  
Row to

$$[A \ \mathbf{v}]$$

it has to  
be consistent

## Bases for Nul A, Col A, and Row A

### THEOREM 6

The pivot columns of a matrix  $A$  form a basis for Col  $A$ .

### Example

Find a basis for Col  $A$

$$A = [a_1 \quad a_2 \quad \dots \quad a_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

the reduced echelon form

$$\begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot columns are C1, C3 and C5.

Therefore,  $\{a_1, a_3, a_5\}$  is a basis of Col  $A$

1, 3, 5 pivot

1. Reduce with RRef

2. find pivot columns

3. these pivots are the basis of Col A

### EXAMPLE 10 Find a basis for the row space of the matrix $A$ from Example 9.

**SOLUTION** To find a basis for the row space, recall that matrix  $A$  from Example 9 is row equivalent to matrix  $B$  from Example 8:

**EXAMPLE 10** Find a basis for the row space of the matrix  $A$  from Example 9.

**SOLUTION** To find a basis for the row space, recall that matrix  $A$  from Example 9 is row equivalent to matrix  $B$  from Example 8:

By Theorem 7, the first three rows of  $B$  form a basis for the row space of  $A$  (as well for the row space of  $B$ ). Thus:

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix} \sim B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

By Theorem 7, the first three rows of  $B$  form a basis for the row space of  $A$  (as well for the row space of  $B$ ). Thus

Basis for Row  $A$  :  $\{(1, 4, 0, 2, 0), (0, 0, 1, -1, 0), (0, 0, 0, 0, 1)\}$

1. Reduce

2. take Rows  $\{(1, 4, 0, 2, 0), (0, 0, 1, -1, 0), (0, 0, 0, 0, 1)\}$  that are basis for Row A

**EXAMPLE 1** Consider a basis  $B = \{\mathbf{b}_1, \mathbf{b}_2\}$  for  $\mathbb{R}^2$ , where  $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  and  $\mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ . Suppose an  $\mathbf{x}$  in  $\mathbb{R}^2$  has the coordinate vector  $[\mathbf{x}]_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ . Find  $\mathbf{x}$ .

**SOLUTION** The  $B$ -coordinates of  $\mathbf{x}$  tell how to build  $\mathbf{x}$  from the vectors in  $B$ . That is,

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2+3 \\ 0+6 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2+3 \\ 0+6 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

See Figure 4.

**EXAMPLE 7** Let

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

and  $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ . Then  $B$  is a basis for  $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ . Determine if  $\mathbf{x}$  is in  $H$ , and if it is, find the coordinate vector of  $\mathbf{x}$  relative to  $B$ .

**SOLUTION** If  $\mathbf{x}$  is in  $H$ , then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

The scalars  $c_1$  and  $c_2$ , if they exist, are the  $B$ -coordinates of  $\mathbf{x}$ . Using row operations, we obtain

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus  $c_1 = 2$ ,  $c_2 = 3$ , and  $[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ . The coordinate system on  $H$  determined by  $B$  is shown in Figure 7.

# ch 4 Lesson 2

**EXAMPLE 3** Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

$$a \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix} \rightarrow v_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

Construct a matrix  $A$  whose columns are the given vectors. Then find the basis of  $\text{Col } A$ . those vectors serve as basis for  $H$  (which is  $\text{Col } A$ )

$$A = \begin{bmatrix} 1 & -3 & 6 & 0 \\ 5 & 0 & 0 & 4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 5 \end{bmatrix} \xrightarrow{\text{rref}(A)} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Pivot columns of } A} v_1, v_2, v_4$$

Therefore,  $\{v_1, v_2, v_4\}$  is a basis of  $H$ . Then,  $\dim H = 3$

## The Dimensions of Nul $A$ , Col $A$ , and Row $A$

The **rank** of a matrix is the dimension of the column space

The number of pivot columns (number of pivots)

The **nullity** of a matrix is the dimension of the null space

The number of nonpivot columns (number of free variables)

The dimension of the row space is equal to the rank

Rank + Nullity = total number of columns

Rank (column space)  $\Rightarrow$  pivot column  
Nullity (null space)  $\Rightarrow$  non-pivot column

Rank + Nullity = total # of column

**EXAMPLE 3** Let  $b_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ ,  $b_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$ ,  $c_1 = \begin{bmatrix} -7 \\ 9 \end{bmatrix}$ ,  $c_2 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$ , and consider the bases for  $\mathbb{R}^2$  given by  $B = \{b_1, b_2\}$  and  $C = \{c_1, c_2\}$ .

- Find the change-of-coordinates matrix from  $C$  to  $B$ .
- Find the change-of-coordinates matrix from  $B$  to  $C$ .

**SOLUTION**

- Notice that  $P_{B \leftarrow C}$  is needed rather than  $P_{C \leftarrow B}$ , and compute

$$[b_1 \ b_2 \ c_1 \ c_2] = \begin{bmatrix} 1 & -2 & -7 & -5 \\ -3 & 4 & 9 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & 3 \\ 0 & 1 & 6 & 4 \end{bmatrix} \rightarrow P_{B \leftarrow C} = \begin{bmatrix} 5 & 3 \\ 6 & 4 \end{bmatrix}$$

- By part (a) and property (6) (with  $B$  and  $C$  interchanged),

$$P_{C \leftarrow B} = (P_{B \leftarrow C})^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -3 \\ -6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -3/2 \\ -3 & 5/2 \end{bmatrix}$$

# ch5 Lesson 1

An **eigenvector** of an  $n \times n$  matrix  $A$  is a nonzero vector  $\mathbf{x}$  such that  $A\mathbf{x} = \lambda\mathbf{x}$  for some scalar  $\lambda$ . A scalar  $\lambda$  is called an **eigenvalue** of  $A$  if there is a nontrivial solution  $\mathbf{x}$  of  $A\mathbf{x} = \lambda\mathbf{x}$ ; such an  $\mathbf{x}$  is called an *eigenvector corresponding to  $\lambda$* .<sup>1</sup>

$$A\mathbf{x} = \lambda\mathbf{x}$$

How to determine whether a given scalar  $\lambda$  is an eigenvalue of matrix  $A$

- Determine if the equation  $(A - \lambda I)\mathbf{x} = \mathbf{0}$  has a nontrivial solution
- If a nontrivial solution exists, then  $\lambda$  is an eigenvalue of matrix  $A$

How to determine whether a given vector  $\mathbf{u}$  is an eigenvector of matrix  $A$

- Calculate  $A\mathbf{u} = \mathbf{b}$
- If vector  $\mathbf{b}$  is a multiple of  $\mathbf{u}$ , then  $\mathbf{u}$  is an eigenvector of matrix  $A$

Given an eigenvalue  $\lambda$ , how to find an eigenvector  $\mathbf{u}$  of a matrix  $A$

- Find the general solution in parametric vector form of the equation:  $(A - \lambda I)\mathbf{x} = \mathbf{0}$
- Each vector in the general solution (assuming any nonzero parameter) is a valid eigenvector of matrix  $A$

How to find the eigenvalues of a matrix  $A$

Find the solution set of the characteristic equation  $\det(A - \lambda I) = 0$

**Special case:** if  $A$  is **triangular**, then the eigenvalues are the entries on the main diagonal.

**EXAMPLE 2** Let  $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ ,  $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$ , and  $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ . Are  $\mathbf{u}$  and  $\mathbf{v}$  eigenvectors of  $A$ ?

This vector is a multiple of  $\mathbf{u}$ .

$$\begin{aligned} & \overset{A}{\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}} \overset{\mathbf{u}}{\begin{bmatrix} 6 \\ -5 \end{bmatrix}} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} \Rightarrow -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} \\ & \qquad \qquad \qquad \div \mathbf{u} \\ & \overset{A}{\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}} \overset{\mathbf{v}}{\begin{bmatrix} 3 \\ -2 \end{bmatrix}} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \Rightarrow \text{X} \\ & \qquad \qquad \qquad \text{Doesn't work} \end{aligned}$$

$\mathbf{u}$  is a eigenvector

**EXAMPLE 3** Show that 7 is an eigenvalue of matrix  $A$  in Example 2, and find the corresponding eigenvectors.

$$J = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Ax = 7x$$

$$Ax - 7x = 0$$

$$(A - 7I)x = 0$$

$$\downarrow$$
$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} =$$

$$\begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = x_2$$

$$x_2 = \text{free}$$

**EXAMPLE 4** Let  $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ . An eigenvalue of  $A$  is 2. Find a basis for the corresponding eigenspace.

$$A - 2I$$

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \sim$$

$$\begin{bmatrix} 2 & -1 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -0.5 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = \frac{1}{2}x_2 - 3x_3$$

$$x_2 = \text{free}$$

$$x_3 = \text{free}$$

### Example:

Is 5 an eigenvalue of  $A = \begin{bmatrix} 6 & -3 & 1 \\ 3 & 0 & 5 \\ 2 & 2 & 6 \end{bmatrix}$ ?

The number 5 is an eigenvalue of  $A$  if and only if the equation  $(A - 5I)x = 0$  has a nontrivial solution. Form

$$A - 5I = \begin{bmatrix} 6 & -3 & 1 \\ 3 & 0 & 5 \\ 2 & 2 & 6 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 1 \\ 3 & -5 & 5 \\ 2 & 2 & 1 \end{bmatrix}$$

and row reduce the augmented matrix:

$$\left[ \begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 3 & -5 & 5 & 0 \\ 2 & 2 & 1 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 8 & -1 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 0 & -5 & 0 \end{array} \right]$$

At this point, it is clear that the homogeneous system has no free variables. Thus  $A - 5I$  is an invertible matrix, which means that 5 is *not* an eigenvalue of  $A$ .

note if  
there isn't a  
free variable

X not eigenvalue

## ch5 Lesson 2

$$\det(A - \lambda I) = 0$$

A scalar  $\lambda$  is an eigenvalue of an  $n \times n$  matrix  $A$  if and only if  $\lambda$  satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

**EXAMPLE 1** Find the eigenvalues of  $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$ .

$$\begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\det \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} = 0$$

$$\left( (2-\lambda)(-6-\lambda) \right) - (3 \times 3)$$

$$-12 - 2\lambda + 6\lambda + \lambda^2 - 9$$

$$\lambda^2 + 4\lambda - 21 - 9$$

$$\lambda^2 + 4\lambda - 21 = 0$$

mode 5.3

$$(\lambda - 3)(\lambda + 7)$$

$$\begin{aligned} \lambda &= 3 \\ \lambda &= -7 \end{aligned}$$

# Similarity 0

**Definition** A  $K \times K$  matrix  $A$  is said to be similar to another  $K \times K$  matrix  $B$  if and only if there exists an invertible  $K \times K$  matrix  $P$  such that

$$B = P^{-1}AP$$

## THEOREM 4

If  $n \times n$  matrices  $A$  and  $B$  are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

# ch5 Lesson 3

## Diagonalization

**EXAMPLE 3** Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix  $P$  and a diagonal matrix  $D$  such that  $A = PDP^{-1}$ .

$$\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} = 0$$

$$R_3 = R_2 + R_3$$

$$\begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 0 & -2-\lambda & -2-\lambda \end{bmatrix}$$

$$C_3 = C_3 - C_2$$

$$\begin{bmatrix} 1-\lambda & 3 & 0 \\ -3 & -5-\lambda & 2+\lambda \\ 0 & -2-\lambda & 0 \end{bmatrix}$$

$$(-1)^{2+3} (2+\lambda) \begin{bmatrix} 1-\lambda & 3 \\ 0 & -2-\lambda \end{bmatrix}$$

$$\begin{matrix} \text{new } c_3 \\ 0 \\ 2+\lambda \\ 0 \end{matrix} = \begin{bmatrix} 3 & -3 \\ -3 & -(-5-\lambda) \\ -2-\lambda & -(-2-\lambda) \end{bmatrix} \rightarrow -3 + 5 + \lambda$$

$$(-2-\lambda) \quad (1-\lambda)(-2-\lambda)$$

$$\frac{(1-\lambda)(-2-\lambda)^2}{-1} = 0$$

$$\downarrow$$

$$(\lambda-1)(\lambda+2)^2$$

$$\lambda = 1$$

$$\lambda = -2$$

$$(A - \lambda I)$$

$$\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

$$\begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \begin{bmatrix} 0 & 1 & 1 \\ -1 & -2 & -1 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 + R_2 = R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\lambda = 1, \lambda = -2, \lambda = -2$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$P D P^{-1} = A \quad \checkmark$$

**EXAMPLE 5** Determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

→ Because all the numbers are different it is diagonalizable

## ch5 Lesson 9

**EXAMPLE 2** Suppose  $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$  is a basis for  $V$ . Let  $T : V \rightarrow V$  be a linear transformation with the property that

$$T(\mathbf{b}_1) = 3\mathbf{b}_1 - 2\mathbf{b}_2 \quad \text{and} \quad T(\mathbf{b}_2) = 4\mathbf{b}_1 + 7\mathbf{b}_2$$

Find the matrix  $M$  for  $T$  relative to  $\mathcal{B}$ .

$$(T(\mathbf{b}_1)) = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$(T(\mathbf{b}_2)) = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \end{bmatrix}$$

# ch5 Lesson 5

power method

(using iteration)  $A = \begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix}$   $x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

| k      | 0                                                                                                                                    | 1                                                                                                                                           | 2                                                                                                                                               | 3 | 4 | 5 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| $x_k$  | $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$                                                                                               | $\begin{bmatrix} 6 \\ 8 \end{bmatrix} \div 8$<br>$\begin{bmatrix} 0.75 \\ 1 \end{bmatrix}$                                                  | $\begin{bmatrix} 1 \\ 0.65 \end{bmatrix}$                                                                                                       |   |   |   |
| $Ax_k$ | $\begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix}$<br>$\begin{bmatrix} 6 \\ 8 \end{bmatrix}$ | $\begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix} \times \begin{bmatrix} 0.75 \\ 1 \end{bmatrix}$<br>$\begin{bmatrix} 11.5 \\ 11 \end{bmatrix}$ | $\begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0.65 \end{bmatrix}$<br>$\begin{bmatrix} 12.69 \\ 12.78 \end{bmatrix}$ |   |   |   |
| $M_k$  | $\begin{matrix} 6 > 8 \times \\ 8 > 6 \checkmark \end{matrix}$<br>8                                                                  | 11.5                                                                                                                                        | 12.78                                                                                                                                           |   |   |   |

1. □□□□

Best to write the exact decimal points

# THEOREM 9

Let  $A$  be a real  $2 \times 2$  matrix with a complex eigenvalue  $\lambda = a - bi$  ( $b \neq 0$ ) and an associated eigenvector  $\mathbf{v}$  in  $\mathbb{C}^2$ . Then

$$A = PCP^{-1}, \text{ where } P = [\operatorname{Re} \mathbf{v} \quad \operatorname{Im} \mathbf{v}] \text{ and } C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

**EXAMPLE 7** Let  $A = \begin{bmatrix} .5 & -.6 \\ .75 & 1.1 \end{bmatrix}$ ,  $\lambda = .8 - .6i$ , and  $\mathbf{v}_1 = \begin{bmatrix} -2 - 4i \\ 5 \end{bmatrix}$ , as in Example 2. Also, let  $P$  be the  $2 \times 2$  real matrix, described in Theorem 9,

$$P = [\operatorname{Re} \mathbf{v}_1 \quad \operatorname{Im} \mathbf{v}_1] = \begin{bmatrix} -2 & -4 \\ 5 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} .8 & -.6 \\ .6 & .8 \end{bmatrix}$$

Check  $A = PCP^{-1} = \frac{1}{20} \begin{bmatrix} -2 & -4 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} .8 & -.6 \\ .6 & .8 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -5 & -2 \end{bmatrix} = \begin{bmatrix} .5 & -.6 \\ .75 & 1.1 \end{bmatrix}$

$P^{-1}$   
 $P$   $C$

①  $P = \begin{bmatrix} -2 & -4 \\ 5 & 0 \end{bmatrix}$

②  $\lambda = .8 - .6i$   $C = \begin{bmatrix} .8 & -.6 \\ .6 & .8 \end{bmatrix}$

$$C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

rule  $\nearrow$

w)  $a^2 + b^2$  on  $P$   
 $0 - (-4 \times 5) = 20$   
 $\frac{1}{20} P C P^{-1}$  (substitute)

# ch6 Lesson 1

## unit vector

length of vector (or norm)  $\|v\|$

The length (or norm) of  $v$  is the nonnegative scalar  $\|v\|$  defined by

$$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}, \text{ and } \|v\|^2 = v \cdot v$$

Let  $u, v$ , and  $w$  be vectors in  $\mathbb{R}^n$ , and let  $c$  be a scalar. Then

- $u \cdot v = v \cdot u$
- $(u + v) \cdot w = u \cdot w + v \cdot w$
- $(cu) \cdot v = c(u \cdot v) = u \cdot (cv)$
- $u \cdot u \geq 0$ , and  $u \cdot u = 0$  if and only if  $u = 0$

$$v \cdot v = v^2$$

$$\|v\| = \sqrt{v \cdot v} = \sqrt{(v_n^2)}$$

(the square cancels Root)

$$\downarrow$$

$$\sqrt{(v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2)}$$

$$\Delta$$

$$\|v\|^2 = v \cdot v = v^2$$

**EXAMPLE 2** Let  $v = (1, -2, 2, 0)$ . Find a unit vector  $u$  in the same direction as  $v$ .

(compute  $v$ )  $\Delta \Delta$

$$\|v\|^2 = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9$$

$$\|v\| = \sqrt{9} = 3$$

$$u = \frac{1}{\|v\|} v$$

$$u = \frac{1}{3} v =$$

$$\frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$

$$\|u\|^2 = u \cdot u = \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + 0^2$$

$$= 1$$

**EXAMPLE 3** Let  $W$  be the subspace of  $\mathbb{R}^2$  spanned by  $\mathbf{x} = (\frac{2}{3}, 1)$ . Find a unit vector  $\mathbf{z}$  that is a basis for  $W$ .

→ (note we have it in fractions)  
we don't want that

(3x) →  $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$$\|y\|^2 = 2^2 + 3^2 = 13$$

$$\|y\| = \sqrt{13} = \sqrt{13}$$

$$\mathbf{z} = \frac{1}{\|y\|} y = \frac{1}{\sqrt{13}} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{bmatrix}$$

# Distance

For  $\mathbf{u}$  and  $\mathbf{v}$  in  $\mathbb{R}^n$ , the **distance between  $\mathbf{u}$  and  $\mathbf{v}$** , written as  $\text{dist}(\mathbf{u}, \mathbf{v})$ , is the length of the vector  $\mathbf{u} - \mathbf{v}$ . That is,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

**EXAMPLE 4** Compute the distance between the vectors  $\mathbf{u} = (7, 1)$  and  $\mathbf{v} = (3, 2)$ .

$\mathbf{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

①  $\mathbf{u} - \mathbf{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$

②  $\|\mathbf{u} - \mathbf{v}\|^2 = (4)^2 + (-1)^2 = 17$

③  $\|\mathbf{u} - \mathbf{v}\| = \sqrt{17}$

**EXAMPLE 5** If  $\mathbf{u} = (u_1, u_2, u_3)$  and  $\mathbf{v} = (v_1, v_2, v_3)$ , then

Same steps

$(\mathbf{u} - \mathbf{v}) =$

$$\begin{bmatrix} u_1 - v_1 \\ u_2 - v_2 \\ u_3 - v_3 \end{bmatrix}$$
$$\|\mathbf{u} - \mathbf{v}\|^2 = (u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2$$
$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

# Orthogonal

Two vectors  $\mathbf{u}$  and  $\mathbf{v}$  in  $\mathbb{R}^n$  are **orthogonal** (to each other) if  $\mathbf{u} \cdot \mathbf{v} = 0$ .

$$\text{if } \mathbf{u} \cdot \mathbf{v} = 0$$

**KAMPLE 1** Show that  $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$  is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = 0$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} = 0$$

$$[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$$

is orthogonal

note: if one of them is not equal to 0 it's not orthogonal

# Orthogonal basis

## THEOREM 5

Let  $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$  be an orthogonal basis for a subspace  $W$  of  $\mathbb{R}^n$ . For each  $\mathbf{y}$  in  $W$ , the weights in the linear combination

$$\mathbf{y} = c_1 \mathbf{u}_1 + \dots + c_p \mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \quad (j = 1, \dots, p)$$

$$\mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \mathbf{u}_j$$

**EXAMPLE 2** The set  $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$  in Example 1 is an orthogonal basis for  $\mathbb{R}^3$ .

Express the vector  $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$  as a linear combination of the vectors in  $S$ .

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

**SOLUTION** Compute

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

$$\mathbf{y} = c_j \mathbf{u}_j$$

$$\mathbf{y} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3$$

$$\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3$$

$$((6 \times 3) + (1 \times 1) + (-8 \times 1)) = 11$$

$$((3 \times 3) + (1 \times 1) + (1 \times 1)) = 11$$

$$\frac{11}{11} \mathbf{u}_1 + \frac{-12}{6} \mathbf{u}_2 = \frac{-33}{33/2} \mathbf{u}_3$$

$$\mathbf{u}_1 - 2 \mathbf{u}_2 - 2 \mathbf{u}_3$$

# ch6 Lesson 2

## Orthonormal Sets

A set of vectors is an **orthonormal set** if all the following are true:

- It is an orthogonal set
- Each vector in the set is a unit vector

A set of vectors is an **orthonormal basis** of a subspace  $W$  if all the following are true:

- It is an **orthonormal set**
- It spans a subspace (not needed if the numbers of vectors in the set is = the dimension of  $W$ )

**EXAMPLE 5** Show that  $\{v_1, v_2, v_3\}$  is an orthonormal basis of  $\mathbb{R}^3$ , where

$$v_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad v_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

check if its  
orthogonal  
( $u \cdot v = 0$ )

$$v_1 \cdot v_2 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} = 0$$

$$v_1 \cdot v_3 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} = 0$$

$$v_2 \cdot v_3 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} = 0$$

it is orthogonal

How to compute orthonormal

$$v_j \cdot v_j = 1$$

$$v_1 \cdot v_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix} \cdot \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix} = 1$$

$$v_2 \cdot v_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} = 1$$

$$v_3 \cdot v_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} = 1$$

(it is an orthonormal set)

# Orthogonal Projections

## THEOREM 8

### The Orthogonal Decomposition Theorem

Let  $W$  be a subspace of  $\mathbb{R}^n$ . Then each  $\mathbf{y}$  in  $\mathbb{R}^n$  can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \quad (1)$$

where  $\hat{\mathbf{y}}$  is in  $W$  and  $\mathbf{z}$  is in  $W^\perp$ . In fact, if  $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$  is any orthogonal basis of  $W$ , then

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \dots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p \quad (2)$$

and  $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ .

**EXAMPLE 2** Let  $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ ,  $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ , and  $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ . Observe that  $\{\mathbf{u}_1, \mathbf{u}_2\}$  is an orthogonal basis for  $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ . Write  $\mathbf{y}$  as the sum of a vector in  $W$  and a vector orthogonal to  $W$ .

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \dots$$

$$\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$$

$$\begin{aligned} \hat{\mathbf{y}} &= \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 \\ \hat{\mathbf{y}} &= \frac{9}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{15}{30} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} \\ 2 \\ \frac{1}{5} \end{bmatrix} \\ \mathbf{z} &= \mathbf{y} - \hat{\mathbf{y}} \\ \mathbf{z} &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -\frac{2}{5} \\ 2 \\ \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{7}{5} \\ 0 \\ \frac{14}{5} \end{bmatrix} \end{aligned}$$

## Ch 6 Lesson 3

# The Gram-Schmidt Process

### THEOREM 11

#### The Gram-Schmidt Process

Given a basis  $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$  for a nonzero subspace  $W$  of  $\mathbb{R}^n$ , define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2$$

$$\vdots$$

$$\mathbf{v}_p = \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 - \dots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}} \mathbf{v}_{p-1}$$

Then  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  is an orthogonal basis for  $W$ . In addition

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} \quad \text{for } 1 \leq k \leq p \quad (1)$$

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$1. \quad \overset{X_1}{\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}}, \overset{X_2}{\begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix}}$$

$$V_1 = X_1$$

$$V_2 = X_2 - \frac{X_2 \cdot V_1}{V_1 \cdot V_1} V_1$$

$$V_1 = X_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

$$V_2 = X_2 - \frac{X_2 \cdot V_1}{V_1 \cdot V_1} V_1 \quad ((8 \times 3) + (5 \times 0) + (-6 \times -1))$$

$$= \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} - \frac{\begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}}{\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}} \cdot \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} =$$

$$((3 \times 3) + (0 \times 0) + (-1 \times -1))$$

$$\begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$$

New basis

$$V_1, V_2$$

$$\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$$

# QR Factorization of Matrices

## THEOREM 12

### The QR Factorization

If  $A$  is an  $m \times n$  matrix with linearly independent columns, then  $A$  can be factored as  $A = QR$ , where  $Q$  is an  $m \times n$  matrix whose columns form an orthonormal basis for  $\text{Col } A$  and  $R$  is an  $n \times n$  upper triangular invertible matrix with positive entries on its diagonal.

$$\begin{matrix} x_1 & x_2 \\ \begin{bmatrix} 1 & 2 \\ 1 & 5 \end{bmatrix} \end{matrix}$$

$$V_1 = X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$V_2 = X_2 - \frac{x_2 \cdot V_1}{V_1 \cdot V_1} V_1$$

$$V_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \frac{7}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} - \begin{bmatrix} \frac{7}{2} \\ \frac{7}{2} \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$$

Length 0

$$v_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{1^2+1^2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$v_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{\frac{-3}{2}^2 + \frac{3}{2}^2}} \begin{bmatrix} -\frac{3}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

(transpose)

$$R = Q^T A$$

$$\begin{bmatrix} \phantom{0} & \phantom{0} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 5 \end{bmatrix}$$

$$= \textcircled{R}$$

just w/ calculator

# ch 6 Lesson 4

## Least-Squares Problems

### DEFINITION

If  $A$  is  $m \times n$  and  $\mathbf{b}$  is in  $\mathbb{R}^m$ , a **least-squares solution** of  $A\mathbf{x} = \mathbf{b}$  is an  $\hat{\mathbf{x}}$  in  $\mathbb{R}^n$  such that

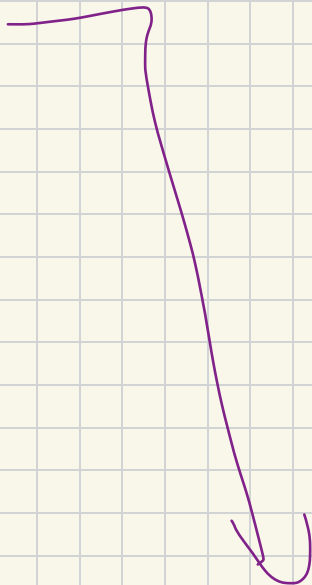
$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all  $\mathbf{x}$  in  $\mathbb{R}^n$ .

### THEOREM 13

The set of least-squares solutions of  $A\mathbf{x} = \mathbf{b}$  coincides with the nonempty set of solutions of the normal equations  $A^T A \mathbf{x} = A^T \mathbf{b}$ .

$$A^T A x = A^T b$$

$\exists x$  

**EXAMPLE 2** Find a least-squares solution of  $A\mathbf{x} = \mathbf{b}$  for

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

$A^T$   $\downarrow$  Calculator

$$A^T A =$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$$

$$A^T \mathbf{b} =$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}$$

Aug matrix  
 $A^T A \mathbf{x} = A^T \mathbf{b}$

R Ref

$$= \begin{bmatrix} 6 & 2 & 2 & 2 & 4 \\ 2 & 2 & 0 & 0 & -4 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & -5 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ -5 \\ -2 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -1 \\ -1 \\ 0 \end{bmatrix}$$

## THEOREM 14

Let  $A$  be an  $m \times n$  matrix. The following statements are logically equivalent. QIMAT

- The equation  $Ax = b$  has a unique least-squares solution for each  $b$  in  $\mathbb{R}^m$ .
- The columns of  $A$  are linearly independent.
- The matrix  $A^T A$  is invertible.

When these statements are true, the least-squares solution  $\hat{x}$  is given by

$$\hat{x} = (A^T A)^{-1} A^T b \quad (4)$$

**EXAMPLE 1** Find a least-squares solution of the inconsistent system  $Ax = b$  for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

$\hat{x}$

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$A^T A x = A^T b$$

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$(A^T A)^{-1} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

# ch 6 Lesson 5